

UCH716: FOOD ENGINEERING AND SCIENCE

L	T	P	Cr
3	1	0	3.5

Course Objective: To impart knowledge to the students about food process engineering, preservation, packaging, related hazards and safety.

Introduction: General aspects of food industry, Composition of foods, quality and nutritive aspects, Characteristic features of processed and natural food, Mass and energy balance in food processing operation.

Food Rheology: Characteristics of non-Newtonian fluids, Time-independent and time-dependent non-Newtonian fluids, linear viscoelastic fluids.

Thermal Processing: Canning/retort processing – process design and equipments. Equipment design aspects of pasteurizer, sterilizers, evaporators and concentrators, Dryers and their design parameters – tray dryer, spray dryer, fluidized bed dryer.

Food Preservation: Microbial Survivor Curves, thermal death of microorganisms and D, Z and F value calculation, Spoilage probability, Food preservation by dehydration, irradiation, Food preservation by adding preservatives.

Food Production, Packaging and Storage: Process design aspects for liquid foods such as milk and juices. Concentration with thermal and membranes processes, Food packaging & product shelf life, Modified atmosphere and controlled atmosphere storage, Aseptic packaging, Freezing and Thawing calculations

Food laws: Legislation, safety and quality control, HACCP method

Course Learning Outcomes (CLOs):

Upon completion of the course, students will be able to:

1. calculate rheological properties of foods.
2. identify and evaluate various design parameters for equipment involved in thermal processing of food.
3. quantify thermal death of micro-organisms and calculate spoilage probability
4. evaluate effect of food processing and packaging/storage on its quality

Text Books:

1. Potter Norman N., Hotchkiss Joseph, *Food Science*, CBS (2005).
2. Toledo Romeo, *Fundamentals of Food Process Engineering*, CBS (2007).

Reference Books:

1. Potty V.H. and Mulky, M.J., *Food Processing*, Oxford and IBH (1993).
2. Heldman D.R. and Singh R.P., *Food Process Engineering*, Chapman and Hall (1984)

3. *Frazier, Food Microbiology, Tata McGraw Hill, (2007).*

Evaluation Scheme:

S. No.	Evaluation Elements	Weightage (%)
1	MST	30
2	EST	50
3	Sessional (may include Quizzes/Assignments/Tutorials evaluation)	20